

THATO BARUTI

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Scientific ML · Physics-Informed Learning · Surrogate Modelling · HPC Validation

PROFILE

Scientific ML engineer completing an MSc in Computational Fluid Dynamics at Cranfield University, building graph-based, physics-informed, and hybrid surrogate models for aerodynamic design under real validation constraints. My work spans model architecture design, uncertainty calibration, multi-fidelity correction against higher-fidelity CFD, failure-regime auditing, and deployment-facing reporting. I am strongest where machine learning, numerical methods, and engineering judgment meet: building models that are not only accurate, but also traceable, stress-tested, and usable in real simulation-driven surrogate and digital-twin-style workflows.

SELECTED RESEARCH AND ENGINEERING EXPERIENCE

AeroGenic Scientific ML Platform

MSc Research, Cranfield University

2025 – Ongoing

- Built DGCNN, MeshGraphNet-style, polar-structured, and target-wise hybrid surrogate models in PyTorch and PyTorch Geometric for aerodynamic coefficient prediction; best held-out results reached 3.18% for C_L , 2.60% for C_D , and 3.83% for C_{My} , with 2.49% on C_D for the polar model and 2.87% deployment-weighted MAPE for the target-wise champion ensemble.
- Built the evidence harness around the models: geometry-disjoint case-level splits, metadata baselines, slice diagnostics, split-conformal uncertainty calibration, and reliability auditing; locked-model audit proved 210/45/45 unique case splits with zero overlap and Spearman 0.613 between predicted and realised held-out risk.

Multi-Fidelity CFD Validation and Residual Correction

Cranfield University HPC Infrastructure

2025 – Ongoing

- Commissioned an automated OpenFOAM RANS validation pipeline and resolved structural failures across meshing, parallel execution, solver stability, and remote environment setup by encoding the fixes directly into the case generator.
- Managed a 32-case validation campaign with 29 returned CFD cases and 26 strict-pass evidence cases, then built grouped geometry-held-out residual correction layers reducing corrected MAE to 0.0677 for C_L , 0.0104 for C_D , and 0.401 for C_{My} .

Optimisation, Selection, and Deployment-Facing Tools

Cranfield University

2025 – Ongoing

- Implemented a requirement-conditioned Pareto optimisation gate over 63,393 candidate designs, retaining 15,031 feasible lift-band candidates and extracting 1,012 non-dominated solutions across drag, trim, lift overshoot, span, and training-manifold risk.
- Exposed the stack through deployment-facing FastAPI and Gradio tools with auditable candidate selection, confidence scores, exported geometry, and JSON reports for higher-fidelity CFD follow-up.

EDUCATION

MSc Computational Fluid Dynamics

Cranfield University, UK

Oct 2025 – Aug 2026 (Expected)

On track for distinction; selected modules include Data Analysis & Post-Processing (87%), Introduction to Fluid Mechanics & Heat Transfer (87%), Turbulence Modelling, Numerical Methods & HPC, Grid Generation, and Numerical Modelling.

BEng Aerospace Engineering

University of Sheffield, UK

Jun 2025

Awards: University of Sheffield International Undergraduate Scholarship; Botswana Government Top Achievers Scholarship.

TECHNICAL SKILLS

ML and Data	PyTorch; PyTorch Geometric; NumPy; scikit-learn; custom training loops; graph neural networks; surrogate modelling; uncertainty calibration; residual correction; Pareto optimisation
Architectures	DGCNN; MeshGraphNet-style message passing; PINNs; hybrid residual models; FiLM conditioning; polar-structured coefficient models
Engineering	Python; C++; Git; experiment evaluation; slice diagnostics; auditable reporting; FastAPI; Gradio
HPC and CFD	Linux HPC workflows; OpenFOAM; ANSYS Fluent; UCNS3D; PBS; SLURM; MPI-parallel simulation pipelines; ParaView; Tecplot
Domain	Aerodynamics; CFD validation; wing design; simulation-driven optimisation; surrogate and digital-twin-style engineering AI workflows